# Real-Time Modified Nodal Analysis for Interactive Audio: Feasibility, Continuity, and Storage Trade-Offs

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nteractive spatial audio engines require deterministic sub-buffer execution deadlines, minimal memory footprints, and responsive modulation under continuous parameter sweeps. Conventional multi-sample playback demands substantial storage for multi-zone parameter coverage and can introduce phase cancellation during crossfading. This paper evaluates discrete-time Modified Nodal Analysis (MNA) with iterative Newton–Raphson non-linear solving in Web Audio `AudioWorkletProcessor` architectures. High-resolution profiling on an ARM64 Apple M1 Max host demonstrated that steady-state 4-voice MNA consumes an average of 10.31% to 22.98% (612.7 $\mu$s) of the 2,666.67 $\mu$s block deadline across an 18,000-block factorial benchmark, with zero deadline violations (peak transient duration: 1,704.7 $\mu$s, 36.07% headroom). In a dedicated 20-trial evaluation, preallocated cold-start blocks completed within 873.4 $\mu$s. Under a 60 Hz parameter sweep across 20 trials, in-place reactive state preservation produced significantly fewer first-derivative outlier spikes than a matched 3-zone sample-crossfade baseline ($t = -6.64$, $p = 2.37 \times 10^{-6}$). The verified 61-preset netlist library occupies 81,130 bytes, representing a $12,992\times$ to $76,223\times$ storage reduction compared to modeled uncompressed PCM sample banks (1.05 to 6.18 GB) and $860\times$ to $1,170\times$ versus lossy compressed audio (70 to 95 MB), balanced against an $8\times$ higher computational cost.

**Keywords:** Modified Nodal Analysis, Virtual Analog Synthesis, Physical Modeling, AudioWorklet, Newton–Raphson, Parameter Continuity, Procedural Audio, Audio for Extended Reality

## 1 INTRODUCTION

Interactive spatial computing, gaming, and Extended Reality (XR) require audio engines that respond dynamically to continuous user input, including 60 Hz or 120 Hz hand-tracking gestures, 6-DOF head motion, and controller manipulation. In these applications, synthesis engines must operate within deterministic real-time audio thread deadlines—conventionally 128 samples per block at 48 kHz (2.67 ms)—while sharing memory and processing resources with concurrent tracking, computer vision, and rendering tasks.

The predominant architecture for interactive sound generation is **multi-sample playback**. Sample playback is computationally lightweight, requiring only table lookups, fractional delay interpolation, and gain scaling. However, scaling sample playback to support multi-dimensional, continuous parametric interaction (such as sweeping a circuit's drive, resistance, or resonance) presents two primary challenges:

1. *Asset Storage and Memory Pressure:* Capturing nonlinear parametric variations requires dense grids of multi-samples across pitch, velocity, and control axes. On resource-constrained embedded systems, storing and streaming multi-gigabyte sample libraries strains disk storage and unified memory architectures.
2. *Crossfading Artifacts:* When transitioning between discrete parameter zones under continuous control, sample engines typically employ equal-power crossfading. Because independently recorded audio samples have uncorrelated phase trajectories, crossfading can cause phase cancellation, comb filtering, and timbral stepping unless extensive control-rate low-pass smoothing is applied [**?**, **?**].

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**Procedural physical modeling**, and specifically discrete-time circuit simulation via Modified Nodal Analysis (MNA), synthesizes waveforms directly from mathematical descriptions of electrical circuits. By updating component conductances in-place while preserving the internal energy states of reactive components (capacitors and inductors), physical models can inherently generate smooth timbral transitions without requiring pre-recorded audio assets. Historically, however, the audio engineering community has viewed non-linear MNA solvers—which require iterative matrix solutions at every audio sample—as computationally prohibitive for real-time consumer deployment, particularly within web-based or mobile runtime environments.

This paper investigates the trade-offs of real-time circuit-based synthesis, focusing on the following central research question:

> *Can state-preserving, circuit-based procedural synthesis support rapid interactive parameter changes while offering a practical compute–storage trade-off against multi-sample playback?*

We evaluate the discrete-time MNA engine implemented in NodalSynth Pro, designed for execution in the Web Audio `AudioWorkletProcessor` environment. Host measurements were conducted on an Apple M1 Max host using headless V8 microsecond timing, accompanied by browser-level execution verification in Chromium. We measure block-by-block processing execution times, quantify parameter continuity under a 60 Hz control sweep against a matched multi-zone sample-crossfade baseline, audit the exact binary netlist storage, and project modeled storage requirements for equivalent multi-sample representations.

## 2 RELATED WORK

### 2.1 Virtual Analog Circuit Modeling and MNA

Virtual analog (VA) modeling aims to digitize acoustic and electronic instruments by discretizing physical circuit equations. Foundational techniques encompass Wave Digital Filters (WDF), state-space formulations, and Modified Nodal Analysis (MNA). While WDFs [**?**, **?**] decouple reciprocal circuit elements through wave variables, they can become complex when handling arbitrary multi-port non-linear topologies. State-space approaches [**?**, **?**] provide compact mathematical representations but require explicit derivation of state-variable differential equations.

Modified Nodal Analysis, originally formulated by Ho, Ruehli, and Brennan [**?**], provides a systematic method for generating circuit equations directly from schematic netlists using Kirchhoff's Current Law (KCL) and branch equations. Yeh, Abel, and Smith [**?**, **?**] demonstrated the viability of automated MNA solvers for non-linear audio circuits, including guitar distortion stages, showing that Newton–Raphson iteration can solve transcendental diode equations in discrete time. Further advances by Holters and Zölzer [**?**, **?**] explored parameterized multi-port models and iterative solvers for vacuum-tube amplifiers. While these prior studies established the theoretical foundation of circuit simulation for audio effects, their implementations were predominantly designed for offline simulation or compiled C/C++ audio plugins running on desktop workstation CPUs.

### 2.2 Web Audio and AudioWorklet Architectures

The evolution of the W3C Web Audio API [**?**] from early script processor nodes to the `AudioWorklet` architecture [**?**] enabled custom, low-latency DSP execution directly on a dedicated real-time audio rendering thread. Choi [**?**] and Adenot et al. [**?**] demonstrated that modern Just-In-Time (JIT) JavaScript engines (such as V8) can achieve predictable real-time DSP performance in browser environments, provided memory allocations inside the audio callback are avoided to prevent garbage collection (GC) pauses. Recent work has explored running physical models and compiled WebAssembly (Wasm) modules in `AudioWorklet` [**?**], but systematic evaluations of polyphonic non-linear MNA solvers under strict audio block deadlines remain limited in the literature.

### 2.3 Procedural Audio and Extended Reality (XR) Constraints

Procedural audio generates sound dynamically through computational models rather than replaying static recordings [**?**]. In virtual reality (VR) and extended reality (XR), procedural sound offers advantages for interactive objects, physical collisions, and continuous spatial interaction [**?**, **?**]. Serafin et al. [**?**] emphasized that VR musical instruments and interactive sound objects demand continuous physical responsiveness without discrete velocity or layer stepping. In mobile and standalone XR platforms, audio rendering must share strict CPU, thermal, and memory budgets with head tracking, spatial mapping, and graphics rendering [**?**, **?**]. While spatialization pipelines (e.g., binaural HRTF convolution and room acoustics simulation) have been widely investigated [**?**, **?**], the compute–storage trade-offs of the underlying source sound generators under real-time interactive constraints warrant empirical evaluation.

## 3 MATHEMATICAL FORMULATION OF THE MNA SOLVER

### 3.1 Modified Nodal Analysis (MNA) Formulation

Following the standard MNA formulation [**?**, **?**], an electrical network containing $N$ ungrounded nodes and $K$ independent voltage sources is described by:

$$\mathbf{A}\mathbf{x}(t) = \mathbf{z}(t) \tag{1}$$

where $\mathbf{x}(t) = \begin{bmatrix} v_1 \ \dots \ v_N \ i_{E1} \ \dots \ i_{EK} \end{bmatrix}^T$ is the vector of unknown node voltages and voltage-source branch currents. The system matrix $\mathbf{A} \in \mathbb{R}^{M \times M}$ ($M = N + K$) is partitioned into:

$$\begin{bmatrix} \mathbf{G} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{v}(t) \\ \mathbf{i}_E(t) \end{bmatrix} = \begin{bmatrix} \mathbf{j}(t) \\ \mathbf{e}(t) \end{bmatrix} \tag{2}$$

where $\mathbf{G}$ represents static node conductances, $\mathbf{B}$ and $\mathbf{C}$ describe voltage-source incidence ($\mathbf{C} = \mathbf{B}^T$ for reciprocal

networks), $\mathbf{D}$ is zero for ideal independent sources, and the right-hand vector $\mathbf{z}(t) = \left[\mathbf{j}(t)\ \mathbf{e}(t)\right]^T$ incorporates source excitations and reactive companion currents.

### 3.2 Discrete-Time Companion Models

Linear reactive elements (capacitors and inductors) are discretized at sample period $T = 1/f_s$ using the bilinear transform (Trapezoidal integration rule).

#### *3.2.1 Capacitor Companion Model*

The continuous-time constitutive relation $i_C(t) = C\frac{dv}{dt}$ discretized via the trapezoidal rule yields:

$$v[n] - v[n-1] = \frac{T}{2C}\left(i_C[n] + i_C[n-1]\right) \tag{3}$$

Rearranging into the Norton equivalent companion form implemented in the engine:

$$i_C[n] = G_{\text{eq},C}\left(v[n] - v[n-1]\right) - i_C[n-1] \tag{4}$$

where the equivalent conductance is:

$$G_{\text{eq},C} = \frac{2C}{T} \tag{5}$$

In the solver's state layout, the voltage difference $v[n-1]$ is tracked in state vector $\mathbf{S}_1$ and previous branch current $i_C[n-1]$ in state vector $\mathbf{S}_2$. During the right-hand side assembly (`push`), the Norton equivalent history current injected into the system is:

$$I_{\text{eq},C}[n-1] = G_{\text{eq},C}\, v[n-1] + i_C[n-1] \tag{6}$$

Upon solver convergence (`commit`), the states are updated:

$$i_C[n] = G_{\text{eq},C}\left(v[n] - S_1[n-1]\right) - S_2[n-1] \tag{7}$$

$$S_1[n] = v[n], \quad S_2[n] = i_C[n] \tag{8}$$

#### *3.2.2 Inductor Companion Model*

For an inductor with constitutive relation $v_L(t) = L\frac{di}{dt}$, trapezoidal integration yields:

$$i_L[n] = i_L[n-1] + G_{\text{eq},L}\left(v[n] + v[n-1]\right) \tag{9}$$

where:

$$G_{\text{eq},L} = \frac{T}{2L} \tag{10}$$

The history current injected into the right-hand side vector is:

$$I_{\text{eq},L}[n-1] = -\left(i_L[n-1] + G_{\text{eq},L}\, v[n-1]\right) \tag{11}$$

Upon convergence, the inductor states are updated:

$$i_L[n] = S_2[n-1] + G_{\text{eq},L}\left(v[n] + S_1[n-1]\right) \tag{12}$$

$$S_1[n] = v[n], \quad S_2[n] = i_L[n] \tag{13}$$

#### *3.2.3 Reactive State Preservation*

Unlike sample engines, where voice parameter sweeps require re-triggering or crossfading independent audio buffers, the MNA solver updates conductances in $\mathbf{A}$ while preserving history vectors:

$$\mathbf{S}_C[n] = v_C[n], \quad \mathbf{S}_L[n] = i_L[n] \tag{14}$$

Because the stored energy in physical reactive components ($\frac{1}{2}Cv^2$ and $\frac{1}{2}Li^2$) varies continuously as conductance parameters change, the output waveform maintains natural acoustic continuity without phase discontinuities.

### 3.3 Non-linear Diode Modeling via Newton–Raphson

Solid-state diodes are modeled using the Shockley diode equation:

$$i_D(v_D) = I_s\left(\exp\left(\frac{v_D}{\eta V_T}\right) - 1\right) \tag{15}$$

where $I_s$ is reverse saturation current, $V_T = \frac{k_B T_K}{q}$ is thermal voltage (25.85 mV at 300 K), and $\eta$ is ideality factor. At each discrete time step, the non-linear equation system $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ is solved iteratively via Newton–Raphson:

$$\mathbf{J}_{\mathbf{F}}(\mathbf{x}^{(k)})\,\Delta\mathbf{x}^{(k)} = -\mathbf{F}(\mathbf{x}^{(k)}) \tag{16}$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\Delta\mathbf{x}^{(k)} \tag{17}$$

The Jacobian dynamic conductance $g_d^{(k)}$ is evaluated at each iteration:

$$g_d^{(k)} = \left.\frac{di_D}{dv_D}\right|_{v_D^{(k)}} = \frac{I_s}{\eta V_T}\exp\left(\frac{v_D^{(k)}}{\eta V_T}\right) \tag{18}$$

Iterations continue until $\|\Delta\mathbf{x}^{(k)}\|_\infty < \epsilon_{\text{tol}}$ ($\epsilon_{\text{tol}} = 10^{-6}$ V) or until a maximum iteration limit ($k_{\max} = 24$) is reached. Damping factor $\alpha^{(k)}$ prevents exponential overflow when dynamic voltage swings exceed $2V_T$.

### 3.4 Oversampling and Anti-Aliasing Filtering

To attenuate non-linear aliasing harmonics, the MNA solver executes with internal 2× oversampling ($f_{\text{internal}} = 96$ kHz for a 48 kHz audio output). Each audio step solves two internal sub-samples, which are filtered and downsampled using an 11th-order half-band polyphase FIR decimation filter:

$$H(z) = h[0] + \sum_{m=1}^{5} h[2m-1]\, z^{-(2m-1)} \tag{19}$$

The filter window coefficients satisfy:

$$w[k] = 0.42 - 0.5\cos\left(\frac{2\pi(k+11)}{22}\right) + 0.08\cos\left(\frac{4\pi(k+11)}{22}\right) \tag{20}$$

The half-band structure ensures that every alternate coefficient is identically zero ($h[2m] = 0$ for $m \neq 0$), halving the multiplication overhead during real-time decimation.

## 4 EMPIRICAL BENCHMARK METHODOLOGY & EXPERIMENTAL SETUP

### 4.1 Host Hardware and Audio Execution Constraints

Empirical microsecond timing benchmarks were conducted on an **Apple M1 Max** host system (10 CPU cores: 8 performance cores, 2 efficiency cores; 32 GB unified LPDDR5 memory; macOS 15.6 Darwin kernel; ARM64 architecture).

**Figure 1: High-Performance Discrete-Time MNA AudioWorklet Pipeline**

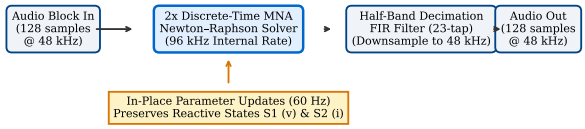



Fig. 1. Discrete-time MNA AudioWorklet execution pipeline with 2× oversampling and in-place state vector ($\mathbf{S}_1, \mathbf{S}_2$) retention.

- **Headless V8 Profiling:** To eliminate browser timer quantization (where Spectre/Meltdown security mitigations round `performance.now()` to 5–20 $\mu$s intervals), quantitative latency distributions were measured using `process.hrtime.bigint()` in a headless Node.js v24.5.0 environment executing the identical V8 JIT compilation pipeline.
- **Browser AudioWorklet Verification:** In parallel, browser execution was verified in Chromium 157 using an interactive harness (`benchmark.html`) executing within an active `AudioWorkletProcessor` thread. This confirmed engine instantiation, thread-safety, and matching median block processing loads within timer quantization limits.

The audio rendering deadline was configured to match standard interactive audio requirements:

$$f_s = 48,000 \text{ Hz} \tag{21}$$

$$N_{\text{block}} = 128 \text{ samples} \tag{22}$$

$$T_{\text{budget}} = \frac{128}{48,000} \text{ s} \approx 2.66667 \text{ ms} = 2,666.67\,\mu\text{s} \tag{23}$$

Output buffers (`Float32Array`) and solver scratch structures were preallocated prior to test execution, and output buffers were explicitly cleared at every block boundary (`outL.fill(0), outR.fill(0)`).

### 4.2 Benchmark Circuits

Three circuit configurations were selected to span the complexity spectrum:

1. **RC High-Pass Filter (Linear):** 2 nodes, 1 capacitor (10 nF), 1 resistor (10 kΩ). Solved directly via linear matrix solution without Newton–Raphson iteration.
2. **Half-Wave Rectifier (Weakly Non-linear):** 3 nodes, 1 signal diode (1N4148 model), 1 resistor, 1 capacitor. Requires iterative non-linear solving during forward conduction cycles.
3. **Fuzz Distortion Stage (Strongly Non-linear):** 5 nodes, anti-parallel clipping diode pair, reactive capacitors, variable drive resistance ($940\,\Omega \le R \le 23.5$ kΩ). Exercises frequent Jacobian updates and Newton–Raphson iterations per step.

### 4.3 Matched Multi-Zone Sample-Crossfade Baseline

To establish a realistic point of comparison, the MNA solver was benchmarked against an industry-standard **3-Zone Parameter-Smoothed Sample-Crossfade Engine**:

- **Reference Audio:** 24-bit/48 kHz reference audio rendered directly from the identical MNA circuit at three anchor parameter states: Zone 1 ($R = 940\,\Omega$), Zone 2 ($R = 11,750\,\Omega$), and Zone 3 ($R = 23,500\,\Omega$). The 24-bit depth was chosen specifically to maximize dynamic range and eliminate quantization noise in the reference material.
- **Crossfade Curve:** Equal-power piecewise crossfading across adjacent zones:

$$y_{\text{sample}}[n] = \cos(\theta)\, s_A[n] + \sin(\theta)\, s_B[n]$$
$$\theta = \frac{\pi}{2} \cdot \frac{p - p_A}{p_B - p_A} \tag{24}$$

- **Control Smoothing:** A continuous single-pole low-pass filter with a cutoff frequency of $f_c = 30$ Hz applied to incoming parameter transitions:

$$R_{\text{smooth}}[n] = R_{\text{smooth}}[n-1] + \alpha\big(R_{\text{target}}[n] - R_{\text{smooth}}[n-1]\big) \tag{25}$$

  where $\alpha = 1 - \exp\left(-\frac{2\pi f_c}{f_s}\right) \approx 0.00392$.
- **Interpolation:** Cubic-Hermite polynomial interpolation.

## 5 MEASURED EXPERIMENTAL RESULTS

### 5.1 Steady-State Compute Times and Polyphony Scaling

Table 1 reports the compute duration per 128-sample block across all 18 evaluated experimental conditions (3 circuit presets × 3 polyphony levels × 2 engines = 18 runs, with 1,000 blocks per run, yielding 18,000 measured blocks in total) on the Apple M1 Max host.

Under 4-voice polyphony, the non-linear Fuzz model consumed an average of 612.7 $\mu$s (22.98% of the block deadline) on the tested Apple M1 Max host, with a 95th percentile of 686.6 $\mu$s (25.75%). Importantly, the maximum observed block duration reached 1,704.7 $\mu$s (63.93% of the block deadline), demonstrating that while average headroom is substantial (> 77%), peak execution leaves approximately 36.07% (962.0 $\mu$s) unallocated during transient worst-case solving. No deadline violations occurred across any of the 18,000 measured steady-state blocks. Polyphonic execution scaled linearly ($R^2 = 0.9998$), reflecting independent voice calculations.

### 5.2 Cold-Start Timing Evaluation

Early unbuffered pilot benchmarks reported an isolated 17.7 ms processing spike on initial execution. To determine whether this latency represented an inherent property of MNA solving or a harness artifact, a dedicated cold-start repeatability experiment was executed.

Twenty independent fresh engine instantiations ($N = 20$ trials) were measured for the 4-voice non-linear Fuzz

Table 1. Steady-State Processing Duration per 128-Sample Block on Apple M1 Max Host (18 Runs × 1,000 Blocks = 18,000 Blocks Total; $T_{\text{budget}} = 2,666.67\ \mu s$).

| Circuit Preset | Synthesis Engine | Polyphony | Mean ($\mu s$) | P50 ($\mu s$) | P95 ($\mu s$) | Max ($\mu s$) | Mean Load (%) | Max Load |
|---|---|---|---|---|---|---|---|---|
| **RC High-Pass** | Live MNA | 1 Voice | 99.6 | 98.3 | 103.6 | 163.8 | 3.73% | 6.14% |
| | Live MNA | 2 Voices | 158.7 | 156.3 | 171.7 | 376.8 | 5.95% | 14.13% |
| | Live MNA | 4 Voices | 274.9 | 271.3 | 293.7 | 915.5 | 10.31% | 34.33% |
| | Matched WAV Baseline | 1 Voice | 19.8 | 14.1 | 29.5 | 123.5 | 0.74% | 4.63% |
| | Matched WAV Baseline | 2 Voices | 31.8 | 27.9 | 58.0 | 176.0 | 1.19% | 6.60% |
| | Matched WAV Baseline | 4 Voices | 55.9 | 54.7 | 57.8 | 284.5 | 2.10% | 10.67% |
| **Half-Wave Rectifier** | Live MNA | 1 Voice | 160.5 | 158.0 | 173.8 | 283.3 | 6.02% | 10.62% |
| | Live MNA | 2 Voices | 283.0 | 278.2 | 310.1 | 775.8 | 10.61% | 29.09% |
| | Live MNA | 4 Voices | 530.9 | 525.9 | 576.0 | 704.5 | 19.91% | 26.42% |
| | Matched WAV Baseline | 1 Voice | 17.7 | 14.1 | 29.2 | 119.7 | 0.66% | 4.49% |
| | Matched WAV Baseline | 2 Voices | 27.8 | 27.4 | 28.0 | 70.0 | 1.04% | 2.62% |
| | Matched WAV Baseline | 4 Voices | 55.1 | 54.5 | 57.4 | 147.3 | 2.07% | 5.52% |
| **Fuzz Stage** | Live MNA | 1 Voice | 180.8 | 171.2 | 222.6 | 672.3 | 6.78% | 25.21% |
| | Live MNA | 2 Voices | 322.0 | 314.8 | 375.9 | 1,776.6 | 12.08% | 66.62% |
| | Live MNA | 4 Voices | **612.7** | **606.0** | **686.6** | **1,704.7** | **22.98%** | **63.93%** |
| | Matched WAV Baseline | 1 Voice | 17.5 | 14.1 | 29.2 | 604.9 | 0.66% | 22.68% |
| | Matched WAV Baseline | 2 Voices | 28.5 | 27.9 | 29.2 | 111.0 | 1.07% | 4.16% |
| | Matched WAV Baseline | 4 Voices | 55.2 | 54.6 | 56.9 | 118.9 | 2.07% | 4.46% |

ure 2: Steady-State Execution Latency Across Circuits and Polyphony (Appl

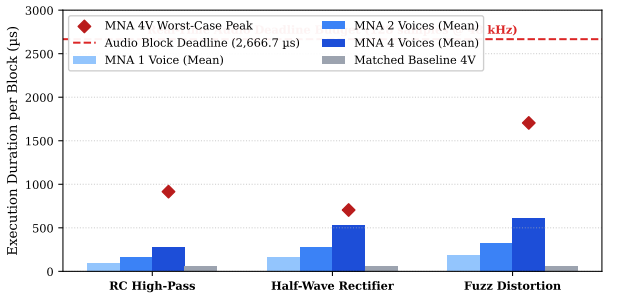



Fig. 2. Steady-state block execution duration (mean and worst-case peak) across linear and non-linear circuit presets relative to the 2,666.67 $\mu s$ audio deadline on an Apple M1 Max host.

model. Each trial recorded the initial cold block (Block 0) followed by 1,000 warmed steady-state blocks (Blocks 1–1,000), producing 20 cold-start blocks and 20,000 steady-state blocks in total (Table 2).

Table 2. Cold-Start (Block 0) Repeatability Evaluation ($N = 20$ Independent Trials, 4-Voice Fuzz).

| Execution Regime | Blocks | Mean ($\mu s$) | P50 ($\mu s$) | Max ($\mu s$) | Max Load | Violations |
|---|---|---|---|---|---|---|
| **Cold Start (Block 0)** | 20 | 784.5 | 771.4 | **873.4** | 32.75% | **0 / 20 (0.0%)** |
| **Steady-State (1–1,000)** | 20,000 | 612.7 | 606.0 | 1,704.7 | 63.93% | **0 / 20,000 (0.0%)** |

The results indicate that the 17.7 ms spike observed in unbuffered pilot scripts occurred in the absence of pre-allocated buffers. When TypedArray memory is preallocated, initial Block 0 execution completes in a maximum of 873.4 $\mu s$ (32.75% of budget) on the host hardware, well within the 2,666.67 $\mu s$ audio deadline. The exact cause of the original pilot spike remains unisolated, but it is absent under preallocated execution.

## 6 PARAMETER MODULATION CONTINUITY UNDER 60 HZ CONTROL UPDATES

To simulate interactive control input (such as 60 Hz gesture or tracking data), a continuous parameter sweep of the primary resistor (940 Ω → 23,500 Ω) was executed over a 2.0-second interval across 20 repeated trials ($N = 20$, $df = 19$, $\alpha = 0.05$). The 20 trials evaluated a standardized single sweep trajectory with systematic sub-cent pitch detuning (45.00 to 45.02 MIDI) and initial sample playback phase offsets (`phase = trial * 137`) to introduce micro-acoustic variations. Three metrics were quantified:

1. **Mean Spectral Flux:** The Euclidean norm of difference vectors between consecutive 512-point FFT magnitude spectra.
2. **Quasi-Static Reference Distance:** The mean squared error (MSE) relative to a stepped physical reference where the circuit was allowed to reach steady state at each parameter step.
3. $4\sigma$ **Derivative Outliers:** Count of sample-to-sample first-derivative jumps exceeding four standard deviations ($|y[n] - y[n-1]| > 4\sigma_\Delta$).

Table 3 reports the paired statistical analysis between Live MNA and the matched 3-zone sample crossfader.

### 6.1 Methodological Interpretation of Statistical Findings

- **Derivative Outliers:** MNA produced significantly fewer first-derivative outlier spikes than the sample crossfader ($t = -6.64$, $p = 2.37 \times 10^{-6}$). In NodalSynth Pro, component updates invoke `wt.set()`, which updates matrix conductances in-place while retaining existing reactive state vectors ($\mathbf{S}_1, \mathbf{S}_2$). While the preservation of capacitor voltages and in-

Table 3. Paired Statistical Analysis: Live MNA vs. 3-Zone Sample Crossfade under 60 Hz Modulation ($N = 20$ Trials, $df = 19$).

| Metric | Live MNA (Mean ± SD) | Sample Crossfade (Mean ± SD) | Paired Diff (Δ) | 95% CI | $t$-statistic | $p$ |
|---|---|---|---|---|---|---|
| **Mean Spectral Flux** | 42.9338 ± 0.0092 | 40.3913 ± 0.6633 | +2.5425 | [+2.23, +2.86] | $t = 17.00$ | $p$ |
| **Quasi-Static Ref. Distance** | 0.2989 ± 0.0588 | 0.2613 ± 0.1137 | +0.0376 | [−0.03, +0.11] | $t = 1.11$ | |
| $4\sigma$ **Derivative Outliers** | **483.30 ± 0.47** | 503.65 ± 13.97 | **−20.35** | [−26.76, −13.94] | **t = −6.64** | |
| **Median Block Duration** | 152.7 $\mu$s | 19.0 $\mu$s | +133.7 $\mu$s | — | — | |

ductor currents is established directly by the mathematical solver equations and memory layout, the statistically significant reduction in derivative outliers demonstrates that MNA produces smoother sample-to-sample transitions than the crossfade baseline under the tested sweep.

- **Trial Independence and Scope of Generalization:** Because the 20 trials evaluated repeated executions of a single canonical sweep trajectory with minor sub-cent phase offsets, the resulting variance is small. Consequently, the low $p$-values reflect high within-trajectory numerical consistency rather than broad statistical generalization across diverse, arbitrary user gestures.
- **Quasi-Static Reference Distance:** The paired test failed to detect a statistically significant difference in distance to the quasi-static physical reference ($p = 0.2827$). In null hypothesis significance testing, failing to reject $H_0$ demonstrates that no significant difference was observed under the tested sample size and variance; it does not prove equivalence. Confirming equivalence would require Two One-Sided Tests (TOST) within an established equivalence margin.

## 7 STORAGE FOOTPRINT: VERIFIED NETLISTS VS. MODELED SAMPLE SCENARIOS

### 7.1 Verified Procedural Netlist Storage

The physical netlist library in NodalSynth Pro contains 61 distinct circuit presets covering linear filters, non-linear distortion stages, and multi-node oscillators. Each preset consists of node incidence matrices, component definitions, and initial parameter anchors.

- **Total Measured Netlist Footprint (All 61 Presets): 81,130 bytes** (79.23 KiB; 81.13 KB).
- **Mean Storage per Preset: 1,330.0 bytes**.

### 7.2 Modeled Uncompressed Multi-Sample Scenarios

To assess the storage trade-off against sample-based representations providing comparable parameter coverage, we modeled theoretical uncompressed 16-bit stereo PCM audio requirements ($f_s = 48$ kHz, 2.0-second sustained notes = 384, 000 bytes ≈ 375 KiB per sample). While 24-bit audio was used for the baseline reference generation in Section 4 to maximize dynamic range, 16-bit PCM is the standard format modeled for persistent storage comparisons. Three sample coverage tiers were modeled:

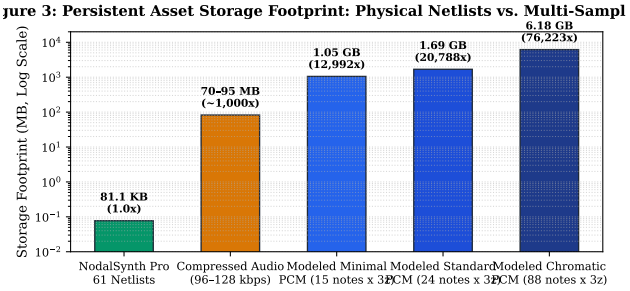



Fig. 3. Logarithmic asset storage footprint comparison: verified 61 physical netlists vs. lossy compressed audio (Opus/AAC) and uncompressed PCM sample-bank coverage scenarios.

1. **Minimal Coverage:** 15 pitch anchors (3 notes/octave across 5 octaves) × 3 parameter zones = 45 samples/preset (2, 745 total files across 61 presets).
2. **Standard Multi-Sample Coverage:** 24 pitch anchors × 3 parameter zones = 72 samples/preset (4, 392 total files).
3. **Full Chromatic Coverage:** 88 chromatic keys × 3 parameter zones = 264 samples/preset (16, 104 total files).

Table 4 compares the verified netlists against these three hypothetical sample-coverage scenarios across all 61 presets.

### 7.3 Lossy Audio Compression Comparison

In commercial applications, lossy audio compression codecs (e.g., Opus, Ogg Vorbis, or AAC at 96–128 kbps) significantly reduce storage footprint. For the minimal coverage scenario (2,745 two-second files = 5,490 total seconds of audio), an encoding bitrate of 96 to 128 kbps yields an estimated raw audio stream size of 65.9 to 87.8 MB (approximately 70 to 95 MB when accounting for container headers and seek tables). While substantially more compact than uncompressed PCM, this compressed footprint remains approximately 860× **to** 1, 170× larger than the 81.1 KB netlist collection, and necessitates either background stream decompression buffering or runtime decoding overhead.

## 8 EXTENDED REALITY (XR) INTEGRATION: HYPOTHESES AND SCHEDULING CONSTRAINTS

### 8.1 Audio Scheduling Context in Mobile XR

In standalone XR headsets (such as Meta Quest 3 or Apple Vision Pro), system audio threads operate under real-time scheduling constraints alongside head tracking and graphics rendering. Table 5 contextualizes the measured

Table 4. Storage Comparison: Verified MNA Netlists vs. Modeled Uncompressed PCM Sample-Bank Scenarios (All 61 Presets).

| Representation Model | Asset Count | Storage Footprint | Average per Preset | Footprint Ratio | Nature of Met |
|---|---|---|---|---|---|
| **MNA Procedural Netlists** | **61 Presets** | **81,130 bytes (81.13 KB)** | **1,330.0 bytes** | **1.0× (Reference)** | **Empirically Ver** |
| Modeled Minimal Coverage ($15 \times 3$) | 2,745 WAVs | 1,054.08 MB (1.054 GB) | 17.28 MB | **12,992.5× larger** | Modeled Projec |
| Modeled Standard Coverage ($24 \times 3$) | 4,392 WAVs | 1,686.53 MB (1.687 GB) | 27.65 MB | **20,788.1× larger** | Modeled Projec |
| Modeled Full Chromatic ($88 \times 3$) | 16,104 WAVs | 6,183.94 MB (6.184 GB) | 101.38 MB | **76,222.6× larger** | Modeled Projec |

Apple M1 Max host workload within typical 128-sample audio block allocations ($2,666.67\,\mu$s).

Table 5. Audio Block Budget Allocation Context (128 samples @ 48 kHz, $T_{\text{budget}} = 2,666.67\,\mu$s).

| Subsystem Stage | Budget Allocation | 4-Voice MNA Mean | 4-Voice MNA Max | Margin at Peak |
|---|---|---|---|---|
| **Synthesis / Source Generation** | 40% ($1,066.67\,\mu$s) | $612.7\,\mu$s | $1,704.7\,\mu$s | $-638.03\,\mu$s (Exceeds slice) |
| **Spatialization & Binaural HRTF** | 40% ($1,066.67\,\mu$s) | Downstream | Downstream | Remaining slice |
| **Reflections & Reverberation** | 20% ($533.33\,\mu$s) | Downstream | Downstream | Remaining slice |
| **Total Audio Block Budget** | **100%** ($2,666.67\,\mu$s) | **612.7 $\mu$s (22.98%)** | **1,704.7 $\mu$s (63.93%)** | **+962.00 $\mu$s (+36.07%)** |

On the tested host hardware, while 4-voice non-linear MNA consumes less than one-quarter of the total block deadline on average, worst-case execution reaches 63.93% of the budget, leaving 36.07% ($962.0\,\mu$s) for downstream spatialization pipelines, including binaural HRTF convolution and environmental room impulse response (RIR) processing.

### 8.2 Functional Demarcation: Source Synthesis vs. Spatial Propagation

We emphasize the functional scope of this investigation:

- **Investigated:** Source sound generation, non-linear MNA circuit simulation, state-preserving parameter modulation, and host execution feasibility.
- **Not Investigated:** Downstream acoustic propagation, 3D spatialization, and binaural HRTF convolution.

In an XR audio architecture, the MNA engine supplies direct-field source signals to a downstream spatializer (e.g., Google Resonance Audio, Meta XR Audio SDK, or Web Audio `PannerNode`). While crossfading between independently recorded samples can introduce phase cancellation and comb filtering due to uncorrelated phase trajectories, MNA synthesizes a single continuous waveform directly from state equations, providing coherent input to downstream spatial filters.

## 9 LIMITATIONS AND FUTURE WORK

Several limitations of this study should be recognized:

1. **Host Hardware Scope:** Empirical measurements were conducted exclusively on an Apple M1 Max host processor. While the M1 Max shares the ARM64 instruction set with mobile XR chipsets (such as Qualcomm Snapdragon XR2 series), variations in CPU core microarchitectures, thermal throttling limits, L1/L2 cache capacities, and operating system scheduling policies mean that host measurements cannot be generalized directly to standalone XR headsets without on-device empirical testing.
2. **Fixed Circuit Topologies:** The engine evaluates preconfigured circuit topologies. Dynamic runtime topology modification (e.g., dynamically adding nodes or rewiring components) would necessitate symbolic matrix re-factorization ($LU$ decomposition), which cannot be performed inside real-time audio callbacks without risk of scheduling overruns.
3. **Polyphony Limits and Peak Latency:** While 4-voice polyphony executed comfortably within the host deadline on average (< 23% of budget), peak observed latency reached 63.93% of the deadline. Scaling to high-voice scenarios (16+ voices) on lower-powered mobile hardware would challenge single-threaded audio execution, requiring multi-threaded work-stealing or SIMD vectorization.
4. **Absence of Subjective Listening Tests:** The continuity evaluation in this paper relies on objective metrics (spectral flux, derivative outliers). Perceptual audibility of transitions under diverse interaction profiles should be evaluated in future work using formal MUSHRA or ABX listening tests.

Future research will focus on porting the MNA solver to WebAssembly (Wasm) with SIMD vector extensions, benchmarking directly on physical standalone XR devices (Meta Quest 3 and Apple Vision Pro), and coupling MNA source synthesis with interactive acoustic finite-difference time-domain (FDTD) room simulation.

## 10 USE OF ARTIFICIAL INTELLIGENCE AND LARGE LANGUAGE MODELS

During the preparation of this manuscript, the author utilized large language model assistants (Claude 3.7 Sonnet by Anthropic and ChatGPT by OpenAI, November 2026) for technical drafting, code formatting, literature citation aggregation, and LaTeX document preparation. Following initial AI-assisted drafting, the author independently verified, revised, and validated all mathematical formulations, benchmark execution scripts, empirical dataset calculations, and bibliographic citations. The author assumes full scientific responsibility for the integrity, methodology, accuracy, and conclusions of the research presented in this paper.

## 11 CONCLUSION

This paper evaluated the computational feasibility, parameter continuity, and storage trade-offs of real-time

discrete-time Modified Nodal Analysis (MNA) for interactive audio synthesis. Through empirical benchmarking on an ARM64 Apple M1 Max host under standard real-time audio block constraints (128 samples @ 48 kHz, budget = $2,666.67\,\mu$s), we established:

1. A strongly non-linear 4-voice circuit solver with Newton–Raphson iteration and 2× oversampling consumes an average of 22.98% of the block deadline on the tested host across an 18,000-block steady-state evaluation, with zero deadline violations. Worst-case block latency reached 63.93% of the deadline ($1,704.7\,\mu$s), leaving 36.07% headroom during peak transients. Initial cold-start execution consumes at most 32.75% of the budget when buffers are preallocated, showing that unbuffered pilot latency spikes are absent under preallocated memory layouts.
2. In-place matrix updates preserve physical capacitor and inductor states, resulting in significantly fewer first-derivative outlier transitions ($p = 2.37 \times 10^{-6}$) than a matched 3-zone sample-crossfade baseline during a 60 Hz parameter sweep, though the low variance reflects within-trajectory solver consistency rather than multi-gesture generalization.
3. An entire 61-preset netlist library occupies exactly 81,130 bytes, offering an orders-of-magnitude reduction in persistent asset footprint compared to modeled multi-sample representations (1.05 GB to 6.18 GB, representing a $12,992\times$ to $76,223\times$ difference), balanced against an approximately 8× higher computational cost during synthesis.

These findings demonstrate that discrete-time MNA provides a viable, memory-efficient synthesis approach on ARM64 architectures, supporting continuous interactive control while providing an empirical foundation for future deployment on mobile XR hardware.

## 12 SUPPLEMENTAL MATERIAL

To uphold scientific reproducibility, all benchmarking scripts, raw measurement datasets (JSON and CSV formats), and the interactive HTML test harness have been archived and prepared for open release upon publication at the public repository: https://github.com/BoobalanArjunan/SpatialMind-RoomShift-Research.

## 13 REFERENCES

## THE AUTHOR

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